Guohan (Harry) Zhang

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EDUCATION

Johns Hopkins University

B.S. in Electrical Engineering; Minor in Applied Mathematics | Expected May 2028

GPA: 3.8/4.0 | **Coursework:** Intro to VLSI, Advanced Digital Systems, Signals & Systems

Fall 2026 (in progress): Robot Sensors/Actuators, Digital Signal Processing.

EXPERIENCE & RESEARCH

Luo Research Lab, Johns Hopkins University | Undergraduate Researcher | Jan 2025–Present

- Fully automated zinc-electrodeposition workflows by integrating OT-2 robotics with ultrasonic actuation, increasing synthesis throughput by 300%.
- Developed Python/NI DAQ controls for waveforms and pressure-valve sequencing across 50+ trials; fully automated X-ray diffraction (XRD) characterization workflows.
- Cut hardware iteration lead time by 80% through CAD design and 3D-printing of custom experimental fixtures and housings in Autodesk Fusion.
- Investigated surface energies and additive adsorption across three zinc facets using VASP density functional theory calculations and Slurm workflows on Rockfish HPC.

Lucid Motors | Manufacturing Engineering Intern | Jun–Aug 2025 | Newark, CA

- Supported electrical engineering work for electric-vehicle manufacturing and tooling projects.

Comillas Pontifical University (ICAI) | Research Student | May–Jul 2026 | Madrid, Spain

- Contributed an adaptive Python algorithm using Gaussian-process uncertainty and bisection to locate neural-network decision boundaries for topology analysis.

Costream (YC S25) | Member of Technical Staff | Dec 2024–Present | Hybrid

- As one of Costream's first five engineers, contributed to Python backend development, acquired research data, and negotiated with external vendors for audio AI R&D.
- Optimized pipelines processing 50,000+ customer interactions for sales analysis; managed technical onboarding for three enterprise pilots.

Southern California Edison | Electrical Engineering Intern (Troy Tech Program) | Jun–Jul 2023 | Chino, CA

- Supported electrical design reviews and field inspections of power-distribution equipment and substations.

SELECTED PROJECTS

CMOS Keyword-Spotting Circuits | Team Project | EN.520.216 Introduction to VLSI | Spring 2026

- Implemented transmission gates, inverters and master–slave DFFs in a digital hierarchy for two 1,024-stage shift registers using Cadence Virtuoso, Spectre and gpdk045.
- Built and simulated 16-tap current-mode pattern banks; designed full/half-strength bias cells and hierarchical subcells for four 1,024-tap banks; integrated threshold/testbench blocks.

Scrawl AI | Creator & Developer | Feb 2026–Present | scrawlai.app

- Built a web/iOS application that extracts handwriting from existing work and renders solved problems in each user's own handwriting; acquired approximately 300 paying customers.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Verilog, SystemVerilog, RISC-V Assembly

Automation & Hardware: NI DAQ, Opentrons OT-2/API, Arduino, LabVIEW, Xilinx FPGA

Design & Research: Cadence Virtuoso, Spectre, CMOS circuit design, DC/transient simulation, Avizo, Autodesk Fusion (CAD), 3D printing, XRD, VASP, Slurm, Rockfish HPC, Linux, Git

LEADERSHIP

Johns Hopkins University

TA, Mastering Electronics (Jan–May 2026); PILOT Leader, Introductory Chemistry (Aug–Dec 2025).